AF3625

Engineering Economics

Subject Outline

School of Accounting and Finance

Hong Kong Polytechnic University

First Semester 2026-27

Mode of Study:	Lecture (26 hours) Tutorial lessons (13 hours)
Assessment:	**Coursework (50%)** **Final Examination (50%)**

Learning Outcomes

Upon completion of the subject, students will be able to:

- understand how the relevant economic factors shape the environment within which an engineering company operates;
- apply the basic cost accounting techniques in the planning and control of engineering and production activities.

Textbook

- *Engineering Economy (18th ed., Global ed.)* by Sullivan, Wicks and Koelling. *Pearson*
- *Foundations of Economics (9th ed.)* by Bade and Parkin. *Pearson*

Teaching Staff

Name	Email	Office	Phone
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Attendance and Class Participation

Students are expected to attend all lectures and tutorial classes. You are also advised to read through the lecture slides and self-study notes before each class for effective learning and active class participation.

Course Contents and Required Readings

Topic 1: Introduction and the Analysis of the Market

The scope of the subject
The economic problem and cost-benefit analysis
The price mechanism – demand, supply and market

Readings: Bade & Parkin, Chapters 1 and 4; Self-study Notes Units 1 & 2 (2.1)

Topic 2: More Analysis of the Market

Consumer surplus, producer surplus and social surplus
Price floor and price ceiling
Elasticities of demand
Tax incidence and demand elasticity

Readings: Bade & Parkin, Chapters 5, 6 and 7; Self-study Notes Unit 2

Topic 3: Cost and Profit

Firm as a production unit
Short-run and long-run decisions
Production and cost in the short run and long run
Economic profit and accounting profit

Readings: Bade & Parkin, Chapter 14; Self-study Notes Unit 3

Topic 4: Market Structure

Profit maximisation
Perfect competition
Monopoly
Efficiency of monopoly relative to perfect competition

Readings: Bade & Parkin, Chapters 15 and 16; Self-study Notes Unit 4

Topic 5: International Trade and Globalisation

Global markets at work
Gains from trade
International trade restrictions

Readings: Bade and Parkin, Chapter 9; Self-study Notes Unit 5

Topic 6: The Time Value of Money

Interest rate and return to capital

Economic equivalence of cash flows

Readings: Sullivan et al., Chapter 4; Self-study Notes Unit 6

Topic 7: Evaluating a Single Project

The present worth method

The future worth method

The annual worth method

The capitalized worth method

Readings: Sullivan et al., Chapter 5; Self-study Notes Unit 7

Topic 8: Comparison and Selection amongst Alternatives

Internal rate of return method

Alternatives with equal lives

Alternatives with unequal lives

Readings: Sullivan et al., Chapters 5 and 6; Self-study Notes Unit 7

Schedule for Lectures

	Thursday 1:30pm-3:20pm (Z207)
Week	**Content**
1	Topic 1
2	Topic 1 and 2
3	Topic 2 and 3
4	Topic 3 and Topic 4
5	***Public Holiday***
6	Topic 4
7	Topic 5
8	Topic 6
9	Topic 6 and 7
10	Topic 7 and 8
11	Topic 8
12	***Revision***
13	***Revision***

Schedule for Tutorial Lessons

Week	*Monday*	*Tuesday*	*Wednesday*	*Thursday*	*Friday*
1	Introduction	Introduction	Introduction	Introduction	Introduction
2	***Self-study notes Unit 1: Self Exercises 1 + Group Formation***	***Self-study notes Unit 1: Self Exercises 1 + Group Formation***	***Self-study notes Unit 1: Self Exercises 1 + Group Formation***	***Self-study notes Unit 1: Self Exercises 1 + Group Formation***	***Self-study notes Unit 1: Self Exercises 1 + Group Formation***
3	***Self-study notes Unit 1: Self Exercises 2 and 3***	***Self-study notes Unit 1: Self Exercises 2 and 3***	***Self-study notes Unit 1: Self Exercises 2 and 3***	***Self-study notes Unit 1: Self Exercises 2 and 3***	***Self-study notes Unit 1: Self Exercises 2 and 3***
4	Group 1 Presentation	Group 1 Presentation	Group 1 Presentation	Group 1 Presentation	Group 1 Presentation
5	Group 2 Presentation	Group 2 Presentation	Group 2 Presentation	***Public Holiday***	Group 2 Presentation
6	Group 3 Presentation	Group 3 Presentation	Group 3 Presentation	Group 2 Presentation	Group 3 Presentation
7	***Self-study notes Unit 2: Self Exercises 4-6***	***Self-study notes Unit 2: Self Exercises 4-6***	***Self-study notes Unit 2: Self Exercises 4-6***	Group 3 Presentation	***Self-study notes Unit 2: Self Exercises 4-6***
8	***Public Holiday***	Group 4 Presentation	Group 4 Presentation	Group 4 Presentation	Group 4 Presentation
9	Group 4 Presentation	Group 5 Presentation	Group 5 Presentation	Group 5 Presentation	Group 5 Presentation
10	Group 5 Presentation	***Self-study notes Unit 3: Self Exercises 1-3***	***Self-study notes Unit 3: Self Exercises 1-3***	***Self-study notes Unit 3: Self Exercises 1-3***	***Self-study notes Unit 3: Self Exercises 1-3***
11	Group 6 Presentation	Group 6 Presentation	Group 6 Presentation	Group 6 Presentation	Group 6 Presentation
12	Group 7 Presentation	Group 7 Presentation	Group 7Presentation	Group 7 Presentation	Group 7 Presentation
13	***Self-study notes Unit 6: Self Exercises 2-4***	***Self-study notes Unit 6: Self Exercises 2-4***	***Self-study notes Unit 6: Self Exercises 2-4***	***Self-study notes Unit 6: Self Exercises 2-4***	***Self-study notes Unit 6: Self Exercises 2-4***

Assessment Method

Assessment of this course is comprised of the following components:

(I) Coursework: 50% of the total assessment marks

- Group Presentation of Tutorial Questions (10%)
- Individual Assignment (15%)
- Mid-Term Test (20%)
- Tutorial Lessons Attendance and Participation (5%)

Details of the Mid-Term Test:

Date and Time	Venue	Topics covered	Format
7th November 2026, Saturday (***Week 10)*** ***5pm – 6pm***	PolyU Campus (exact venue to be announced later)	Topic 1- 4	Multiple choice questions & Problem set type questions

(II) Final Examination: 50% of the total assessment marks

- Date: December 2026 (during the Examination Period)
- Coverage: Topic 1-8

Tutorial Lesson Activities

- There will be either discussions or group presentation in each tutorial class.

(I) Discussions involving the whole class

- All students are required to prepare the answers of the assigned exercises before class and share their answers in the tutorials.

(II) Group presentation

- Seven groups will be formed in each tutorial class. ***Each group will be assigned FOUR presentation questions as specified in the file "Tutorial Presentation Questions".***
- Refer to the tutorial lesson schedule in this subject outline to find out the week of your presentation.
- Each group will present in class the answers of the four assigned questions in the file "Tutorial Presentation Questions".
- Please make sure that the ***workload is distributed equally among members*** and please help your group members in the preparation as well as the presentation in class.
- Group members may approach their tutor before presentation to clarify any doubts about their assigned questions. However, you cannot ask your tutor to verify your answers.
- The ***maximum time limit*** for each group presentation is ***25 minutes***.
- The audience is required to participate actively in each tutorial. Please listen attentively and raise questions or offer comments relating to the presentation in order to enhance the learning of the whole class.
- The marking of your presentation will be based on the following criteria:

▪ Accuracy	35%
▪ Elaboration and details	25%
▪ PowerPoint design	15%
▪ Presentation skills	15%
▪ Teamwork	10%

- Students should not sit in tutorial classes in which he/she has not been enrolled without the prior approval of the tutor. A student will be disqualified (receive grade F for the presentation) if he/she attends a different tutorial class before conducting his/her own presentation.

Individual Assignment:

- Within one week after the presentation, ***each group member has to submit an individual assignment containing the answers to the THREE assigned questions specified in the file "Tutorial Presentation Questions".*** For example, each member of Group 1 has to type the answers of Questions 1.2, 1.3 and 1.4 in his/her individual assignment.
- You must elaborate your written answers thoroughly assuming that your reader has never watched your presentation in class and does not know the answers.
- The number of words in your assignment should be ***600 - 1,000 words*** (NOT including the cover page or diagrams, if any). The assignment should be ***typed*** with one-and-a-half or double-line spacing using font size 12. State the number of words on the cover page of your report. ***5 marks will be deducted for hand-written assignment***.
- Please refer to the last page of the subject outline for the template of the individual assignment's cover page.
- Upload your individual written assignment through TURNITIN in your TUTOR'S Blackboard subject site before the deadline (1 week after your presentation).
- Also, please email your individual written assignment to your tutor by the deadline (1 week after your presentation).
- DO NOT COPY FROM SOMEONE ELSE'S WORK BECAUSE PLAGIARISM IS A CRIME AND WILL BE HEAVILY PENALIZED. If the similarity index is unusually high, your tutor will check if that is a case of plagiarism.

- The marking of your written assignment will be based on the following criteria

▪ Accuracy	50%
▪ Elaboration with necessary details	17%
▪ Structure	17%
▪ Correct English and Economic expressions	16%

- ***The penalty of one mark deduction per day*** will be imposed for late submission of the individual assignment after the deadline.

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Individual Assignment

Name:	
Student number:	
Tutor:	Dr. WONG Tina / Mr. CHAN Szenam, Alan (Please choose one)
Tutorial day & time:	
Group:	
Questions:	
Number of words:	